S ⇒ "111000"

1	1	1	0	0	0

Zero Start →

0	1	0	1	0	1

One Start →

1	0	1	0	1	0

Twist

Type 1 free of Cost

111000 → y

110001 → x

100011 → x, y

000111

001110

011100

⇒ How to handle Type 1 ?

111000

110001

100011

000111

001110

011100

100000

111000 + 111000

1111000

n

1888 → Medium

Minimum Number of Flips to Make the Binary String Alternating :-

s = "111000" —Alternate→ 101010
→ 010101

Rule :- ① :- Remove char start append to last

111000 → 110001 ✗

② :- Pick any idx flip → 0 → 1, 1 → 0

111000

↓

111100

00
11

⇒ Return Minimum type 2

Think ? Agar String to Alternate banani hai → adjacent char same ✗

0, 1

Zero Start :- 0 1 0 1 0 1 0

One Start :- 1 0 1 0 1 0 1

s = "111000"

⇒ First Approach :- String S → S+S

Sare substring nikal lo

Apply logic of Zero/One Compare

⇒ Better :- Instead of Pura ~~Pura~~ Substring nikalna.

back with Window :-

111000111000

0 1 2 3 4 5 6 7 8 9 10 11

Z O

i = 0

j = 0

⇒ Finally First Approach :-

- String S → S+S
- Window se substring ~~nikal~~ get
- Har substring. Apply logic of Zero/One compare

but wait

10^5 → n^2 ✗ TLE

⇒ Optimize Kon sa Kam bar bar Karna pad rahe

Memory

S → S+S ✓

Window → ✓

Har Window Same apply

Zero/One

min

Dry Run :-

s = "100"
n = 3
sb = "100100"

Zero Pattern	0	1	0	1	0	1
	0	1	2	3	4	5
One Pattern	1	0	1	0	1	0

⇒ First Window :-

100100

100

i	sb[i]	Zero Pattern	Mismatch (0)	One Pattern	Mismatch (1)
0	1	0	✓	1	✗
1	0	1	✓	0	✗
2	0	0	✗	1	✓

Count ⇒ Zero Count = 2 ✓
One Count = 1

Ans = 1

Z → 0 1 0 1 0 1
1 0 0 1 0 0]
O → 1 0 1 0 1 0

0 1 0 1 0 1
1 0 0 1 0 0
1 0 1 0 1 0

010
101

Sliding Window :-

Window	i add	i remove	Window Str	Zero Count vs 010	One Count vs 1010	ans
0-2	2	—	100	~~2~~ 1	~~1~~	1
1-3	3	0	001	1	2	1
2-4				~~1~~ 0	~~3~~	0
3-5				1	2	0

Algo / Flow :-

- Double the string
  - S+S to simulate all rotations.
- Create two pattern (Size 2n) :-
  - Zero Pattern → 010101....
  - One Pattern → 101010....
- Use sliding window of size n.
- Count mismatches with both pattern ('0', '1')
- First window :- Calculate mismatch count.
- Start window :-
  - Add mismatch for new char
  - Remove mismatch for old char
- At each step

  ans = min(ans, min(zero '0', One '1'))

Ans ↑

T.C = O(n) → (n)

S.C = ~~O(n)~~ 3n → O(n) + O(n) + (n

(3n)